Untitled

Latent Matrix Factorisation

Factorisation Machine

$$
\begin{aligned}
\hat{y}(\mathbf{x}) &= Pr\Big[Y = 1 \Big| \mathbf{x}\Big] \\
&= w_0 + \sum_{i=1}^{p} w_i x_i + \underbrace{\sum_{i=1}^{p} \sum_{j=i+1}^{p} \langle v_i, v_j \rangle x_i x_j}_{\text{feature interaction block}} \\
w_0 &= \text{" global bias "} \\
v_i &= \text{" latent vector representation of feature } i \\
v_j &= \text{" latent vector representation of feature } j \\
p &= \text{number of features} \\
\langle a, b \rangle &= \text{dot product between } a \text{ and } b
\end{aligned}
$$

Field-Aware Factorisation Machine

Field-aware factorisation machines are almost the same as **factorisation machines**, except that each feature is allowed a different latent factor vector representation within each field.

Example data:

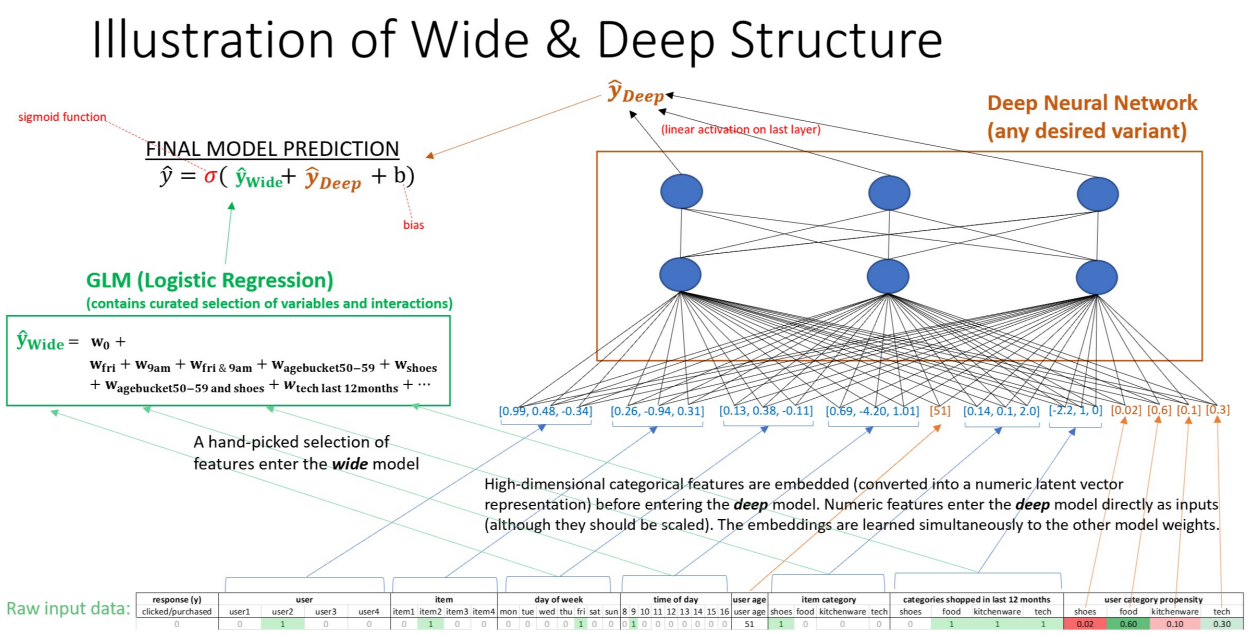
field	feature value
outcome (y)	clicked on item (or could be “bought item”)
user ID	user 1
user ID	user 2
user ID	user 3
item ID	item 1
item ID	item 2
day of week	Monday
day of week	Tuesday
day of week	Wednesday
day of week	Thursday
day of week	Friday
day of week	Saturday
day of week	Sunday
item category	shoes
item category	food
item category	tech

$$
\begin{aligned}
\hat{y}(\mathbf{x}) &= Pr\Big[Y=1\Big|\mathbf{x}\Big] \\
&= w_0 + \sum_{i=1}^{p} w_i x_i + \underbrace{\sum_{i=1}^{p}\sum_{j=i+1}^{p} \langle v_{i,f_2}, v_{j,f_1}\rangle x_i x_j}_{\text{feature interaction block}} \\
w_0 &= \text{" global bias "} \\
v_{i,f_2} &= \text{" latent vector representation of feature } i \text{ in field } f_2 \text{ "} \\
v_{j,f_1} &= \text{" latent vector representation of feature } j \text{ in field } f_1 \text{ "} \\
f_1 &= \text{field of feature } i \\
f_2 &= \text{field of feature } j \\
p &= \text{number of features} \\
\langle a, b\rangle &= \text{dot product between } a \text{ and } b
\end{aligned}
$$

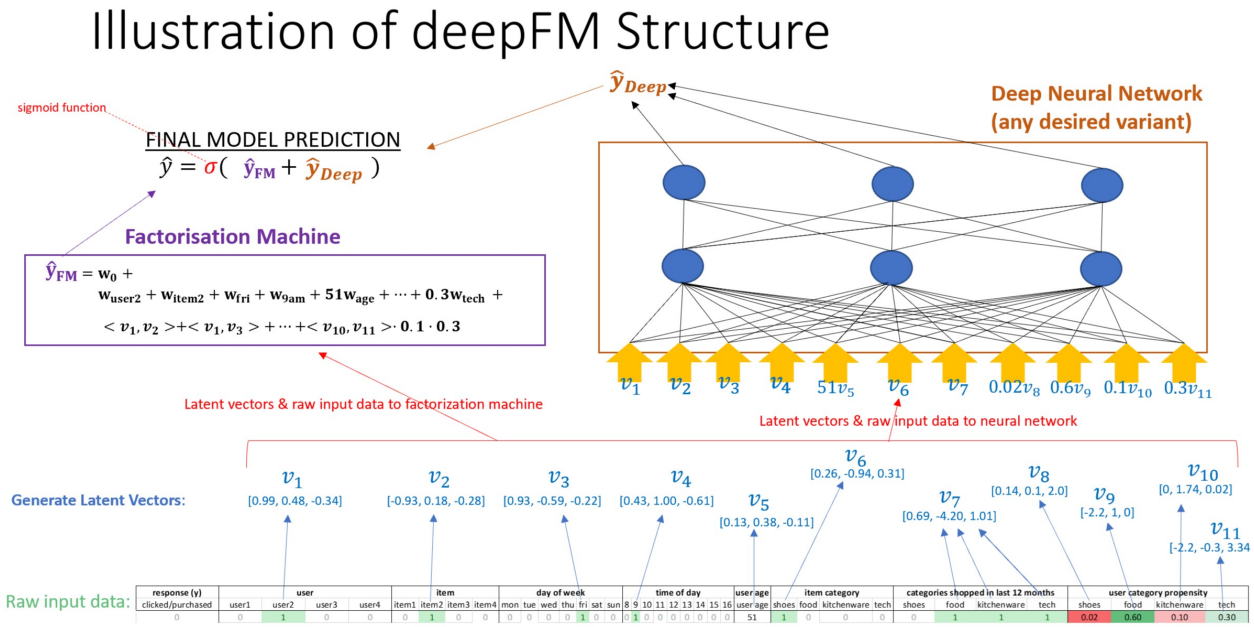
So, for example, a user (e.g. user 69) will have 3 different vector representations - one in each field ({itemID, day of week, item category})

Categorical Feature Embedding

Wide & Deep Network (pre-knowledge for deepFM)



deepFM



References

- DeepFM: An End-to-End Wide & Deep Learning Framework for CTR Prediction (https://arxiv.org/pdf/1804.04950.pdf)